

图 4.1-2

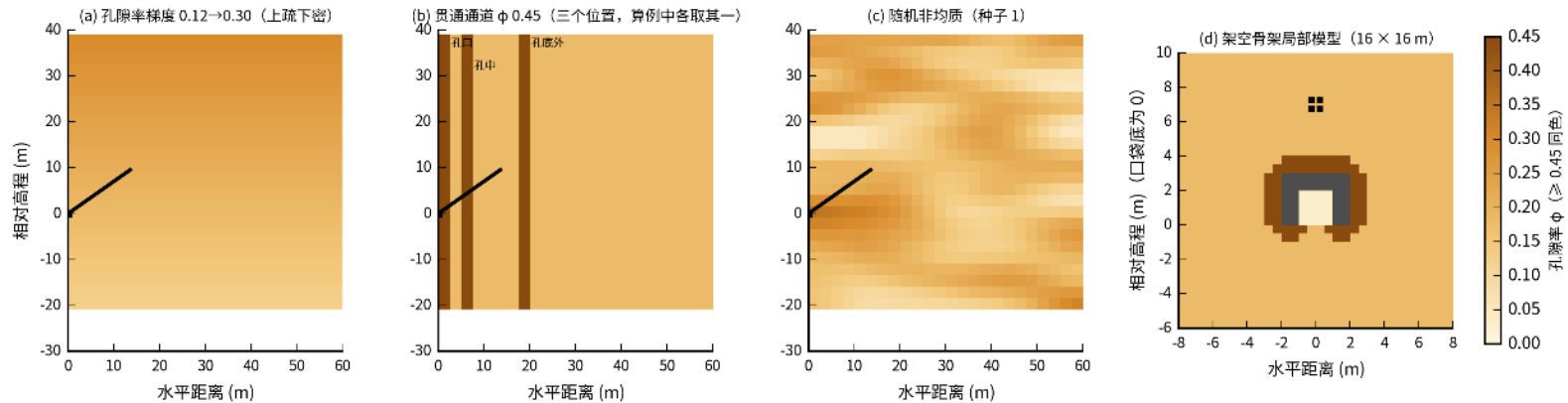


图 4.3-1

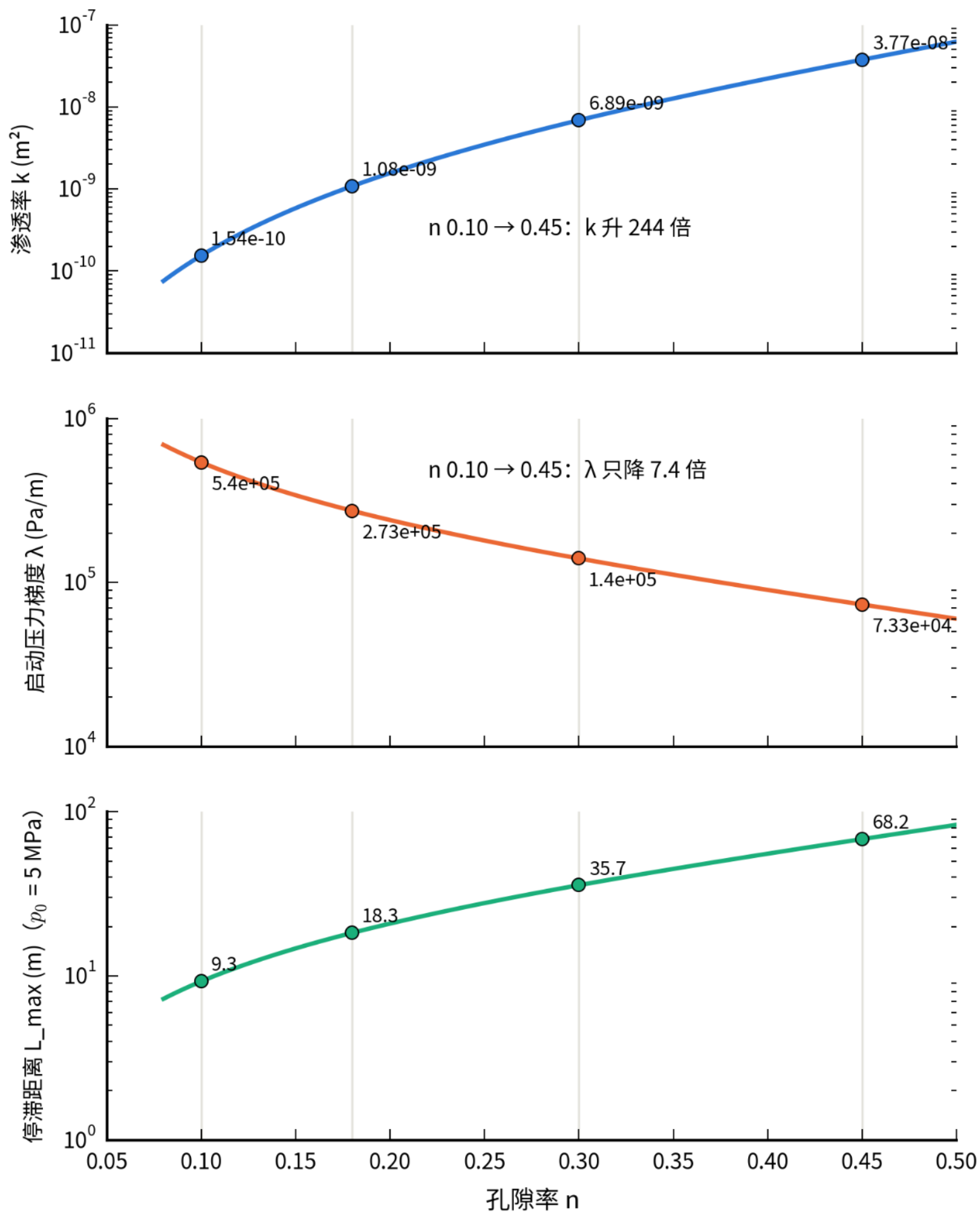


图 4.4-2

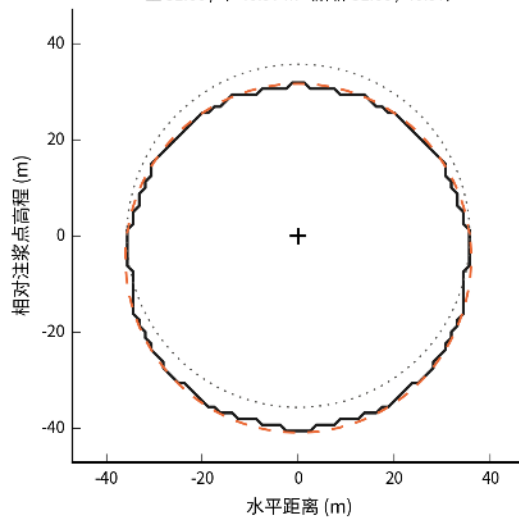
(a) 一维变饱和和充填：锋面位置—时间
(台阶 2b, 对 Gustafson-Stille 解析曲线)

数据待补
(仓库未保存, 需运行验证阶梯测试夹具)

(b) 径向停滞：沿轴压力剖面
(台阶 3, 对 $\max(0, p_0 - \lambda r)$ 解析线)

数据待补
(仓库未保存, 需运行验证阶梯测试夹具)

(c) 均质点源可达域与解析椭圆 ($n = 0.30$)
上 31.66 / 下 40.97 m (解析 31.66 / 40.97)



— 可达域 (二维, 16 邻域)
- - - 解析椭圆 $r = L_{\max} / (1 + n_g \sin \theta)$
... 无重力解析圆

图 4.5-1

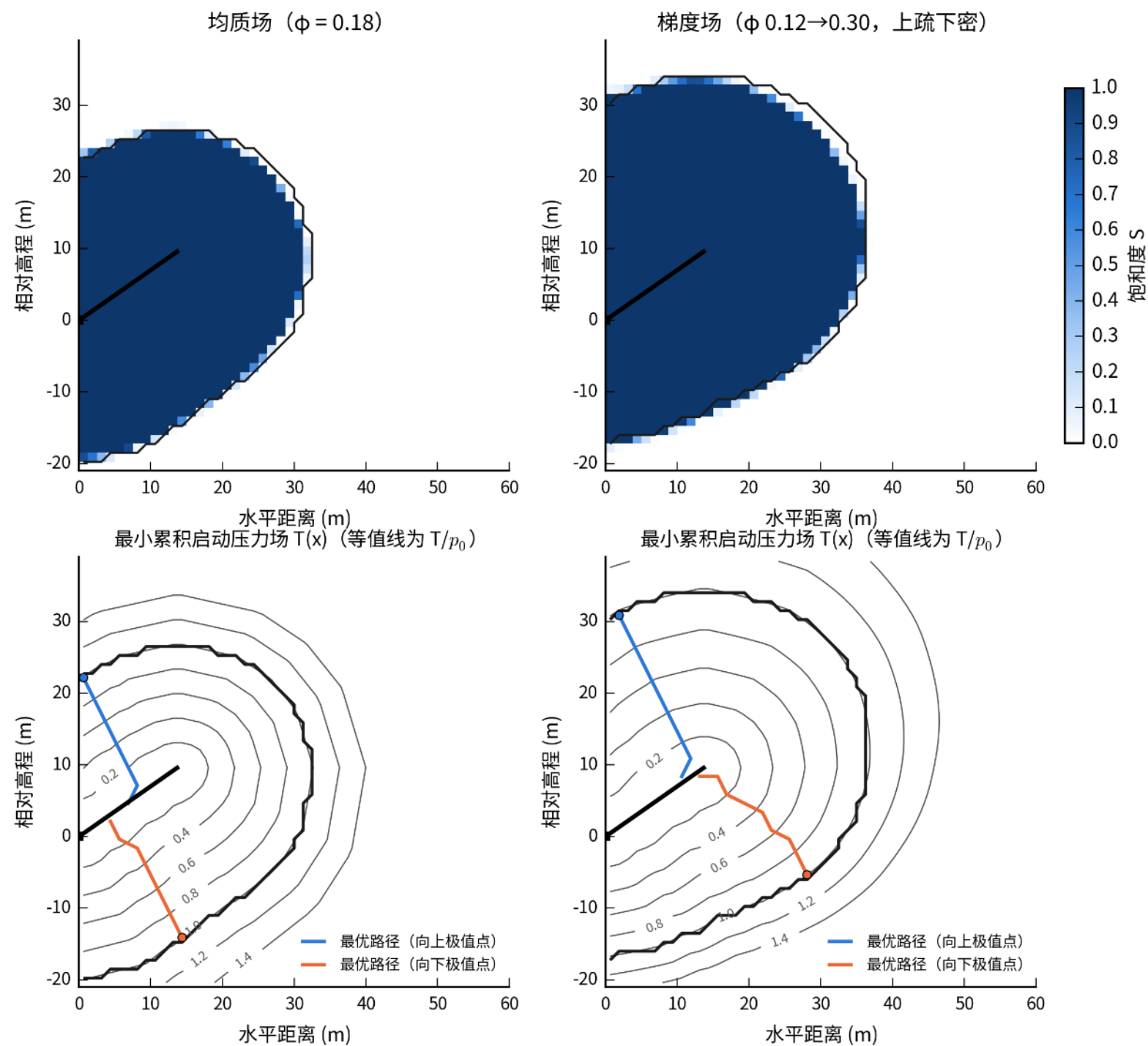


图 4.5-2

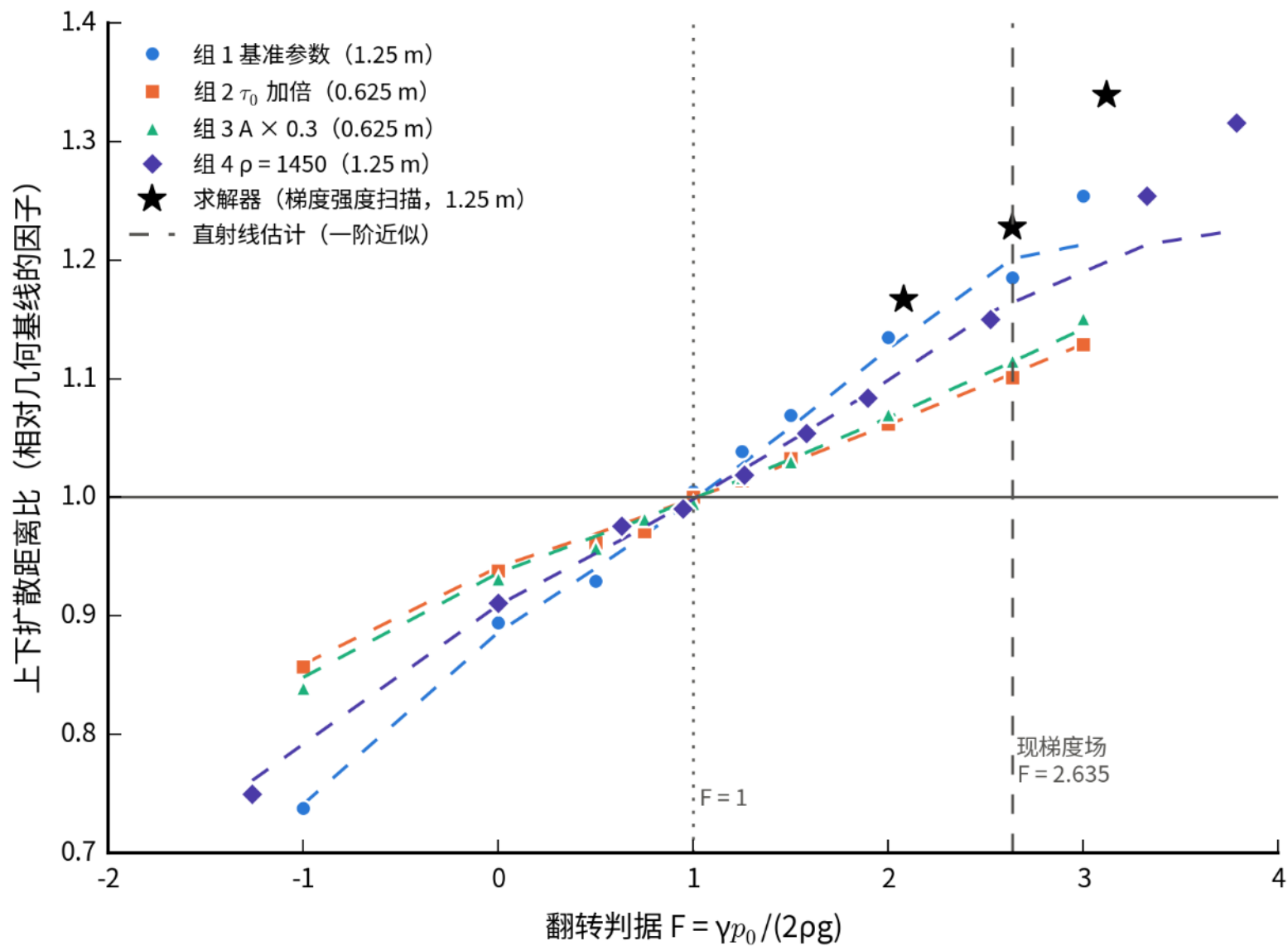


图 4.5-3

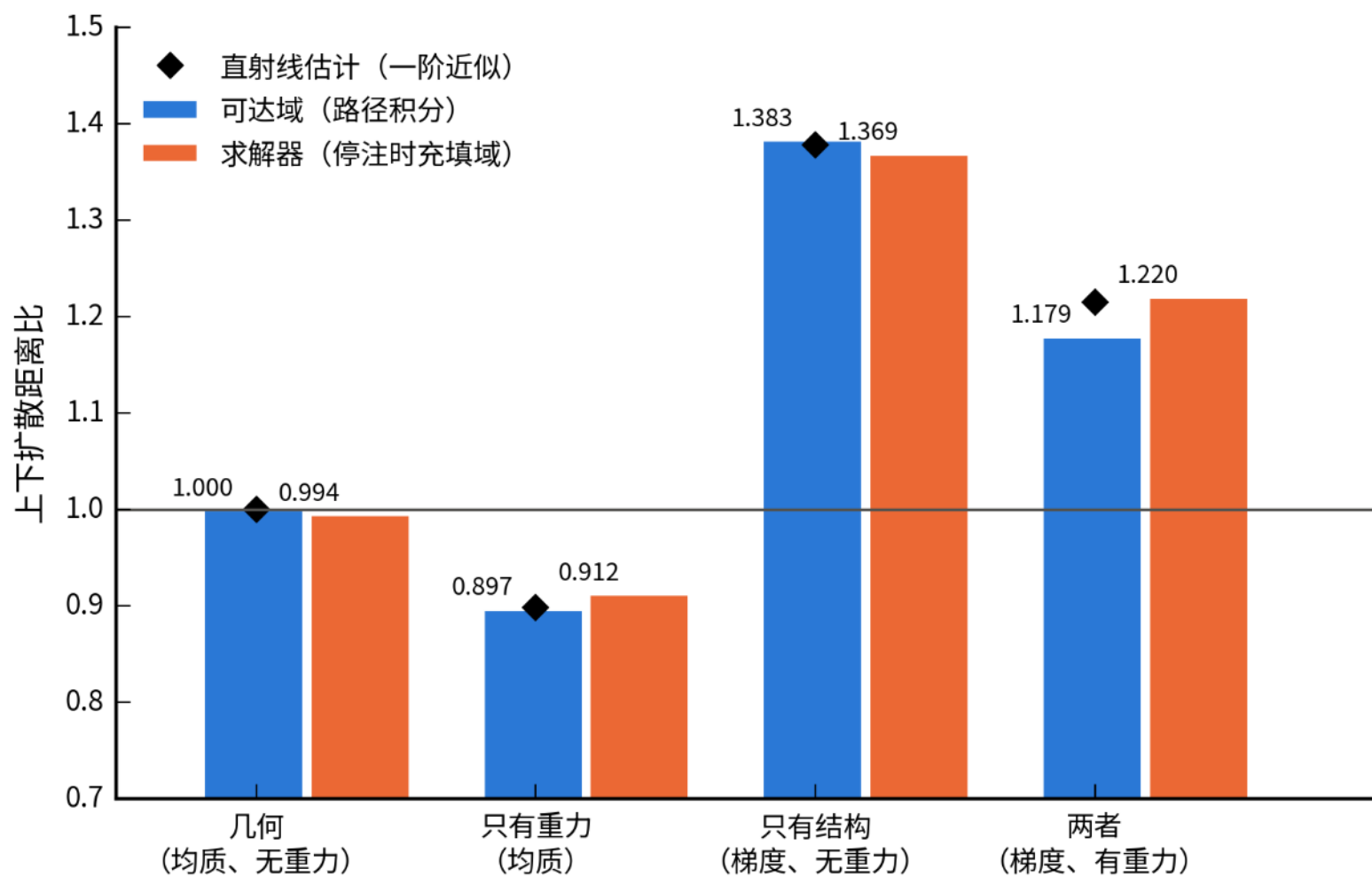


图 4.5-4

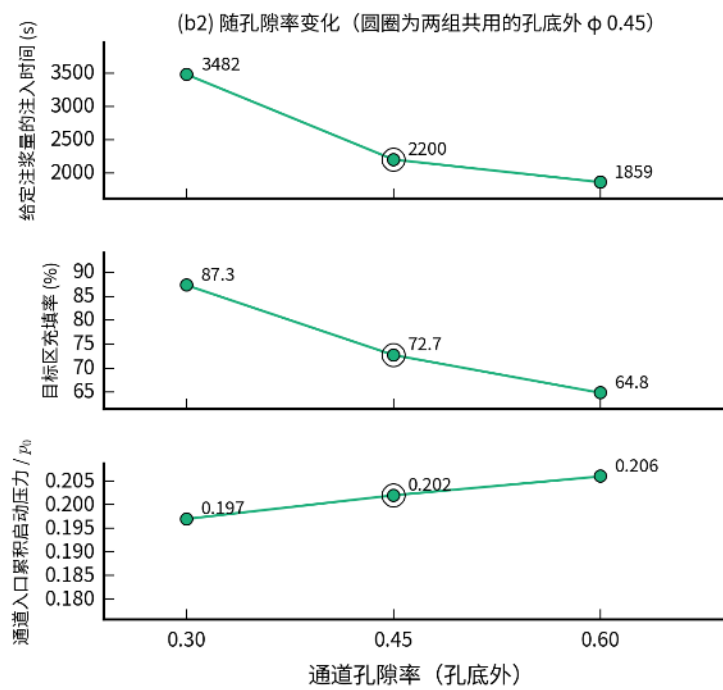
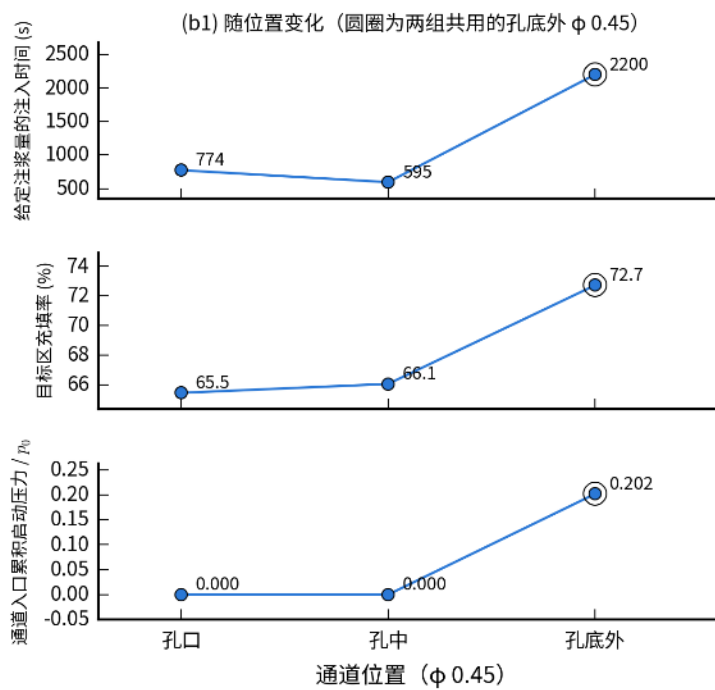
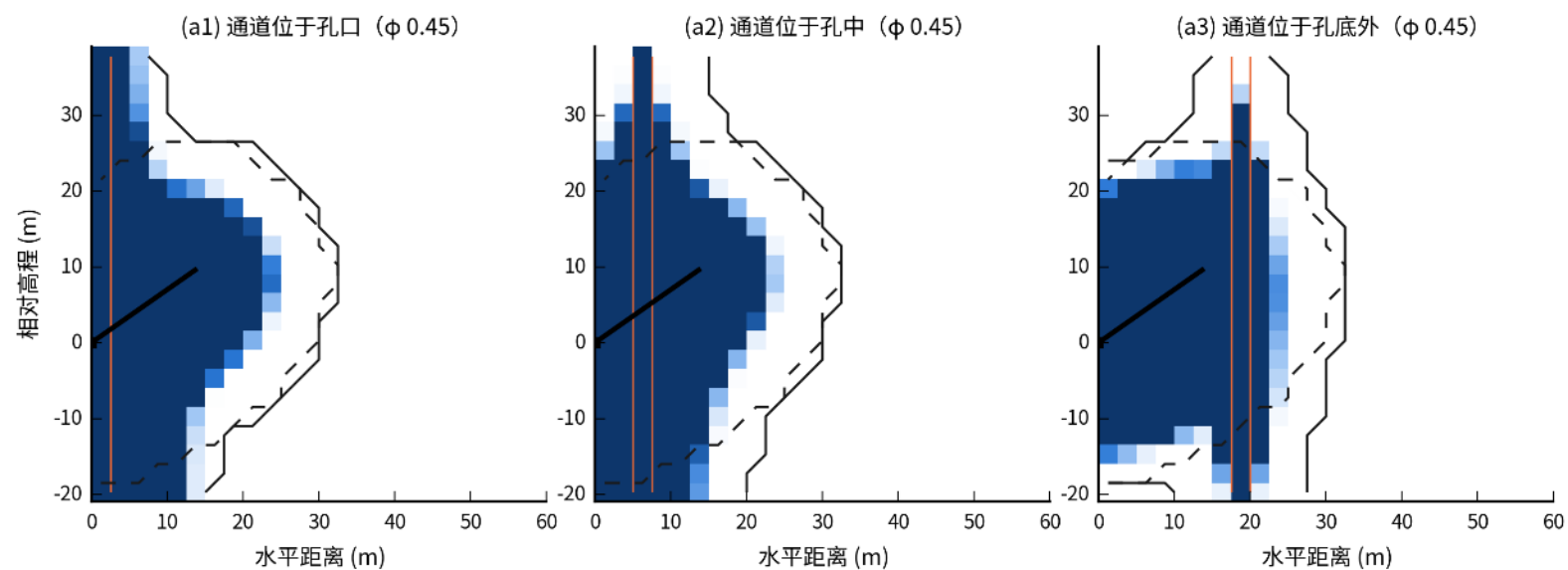


图 4.5-5

设计域未充填部分的成因（不可达区 = 设计缺陷，锋面截断区 = 充填缺陷）

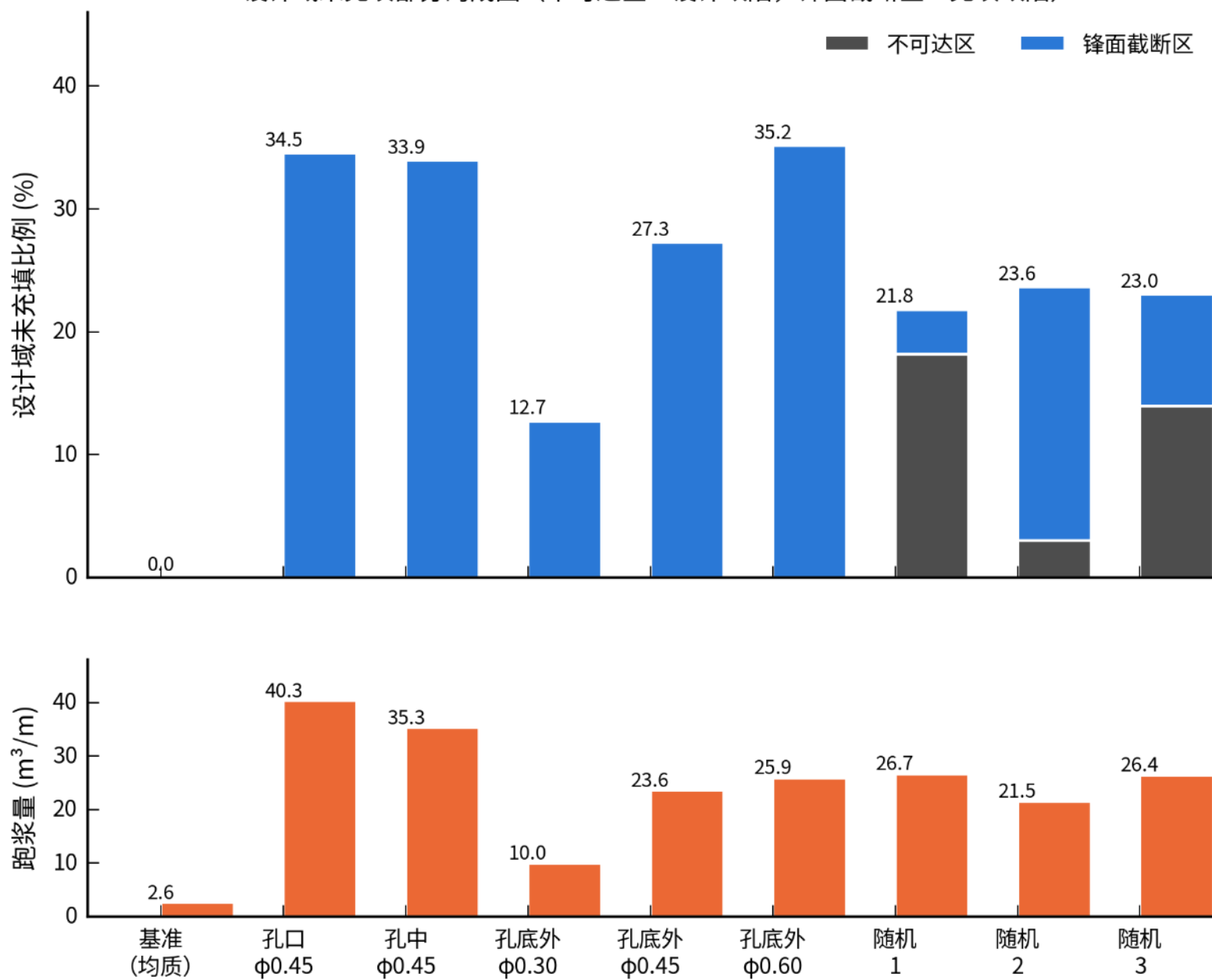
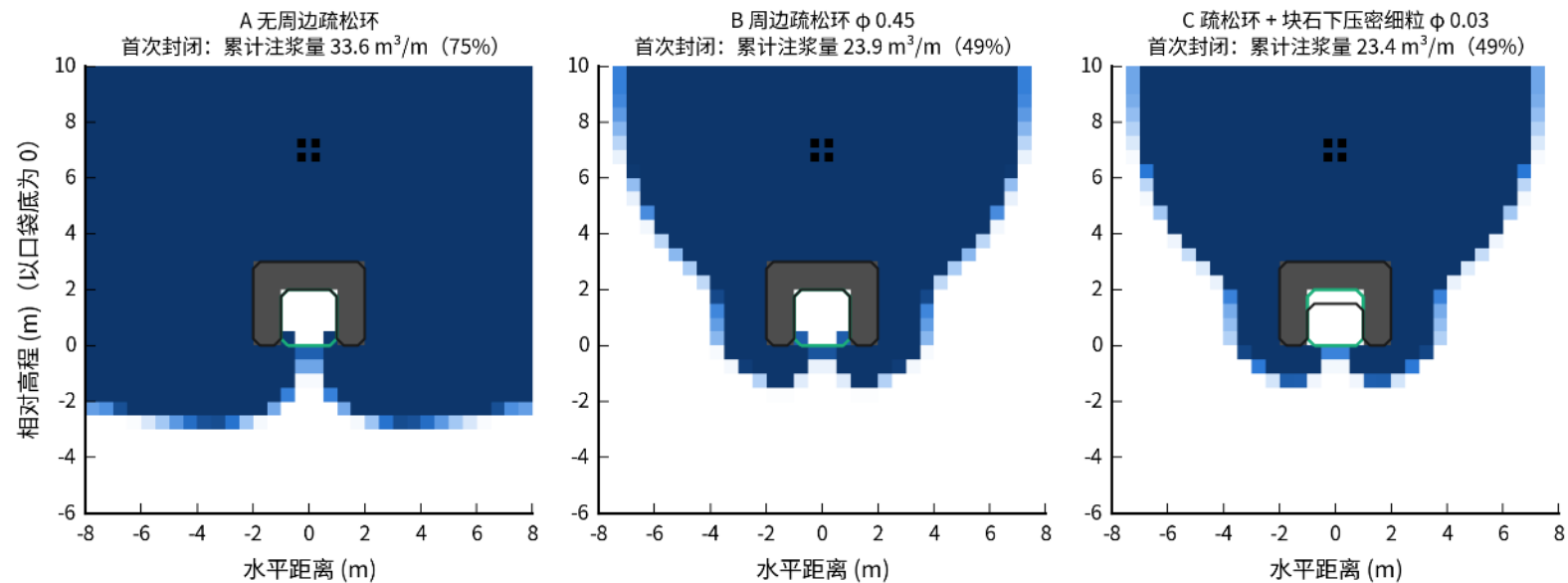


图 4.5-6



色斑: 充填域 (首次封闭时); 深灰: 块石; 绿线: 块石下口袋; 实线: 可达域; 黑方块: 注浆源

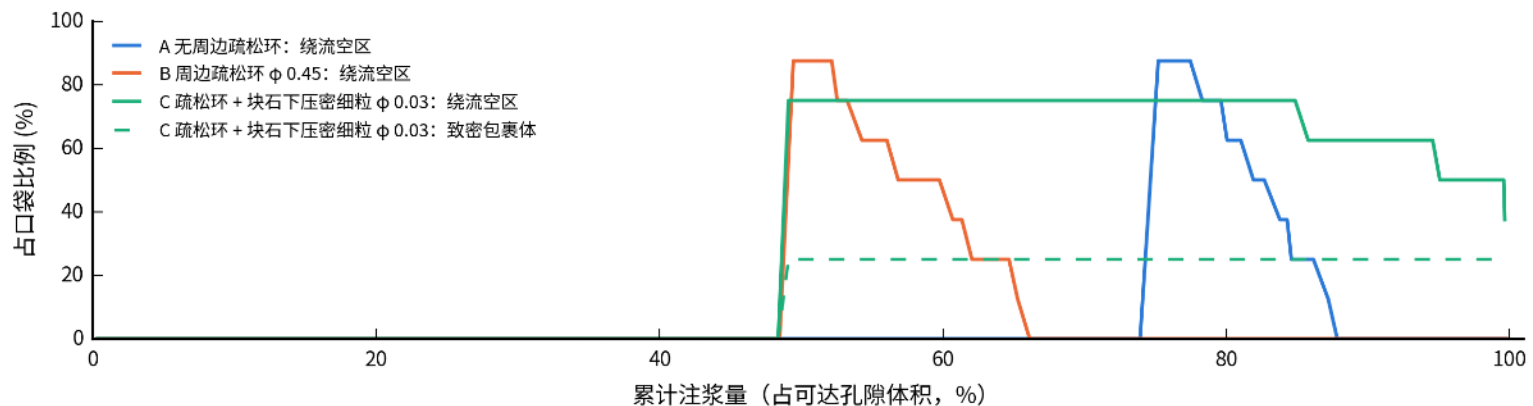


图 4.6-2

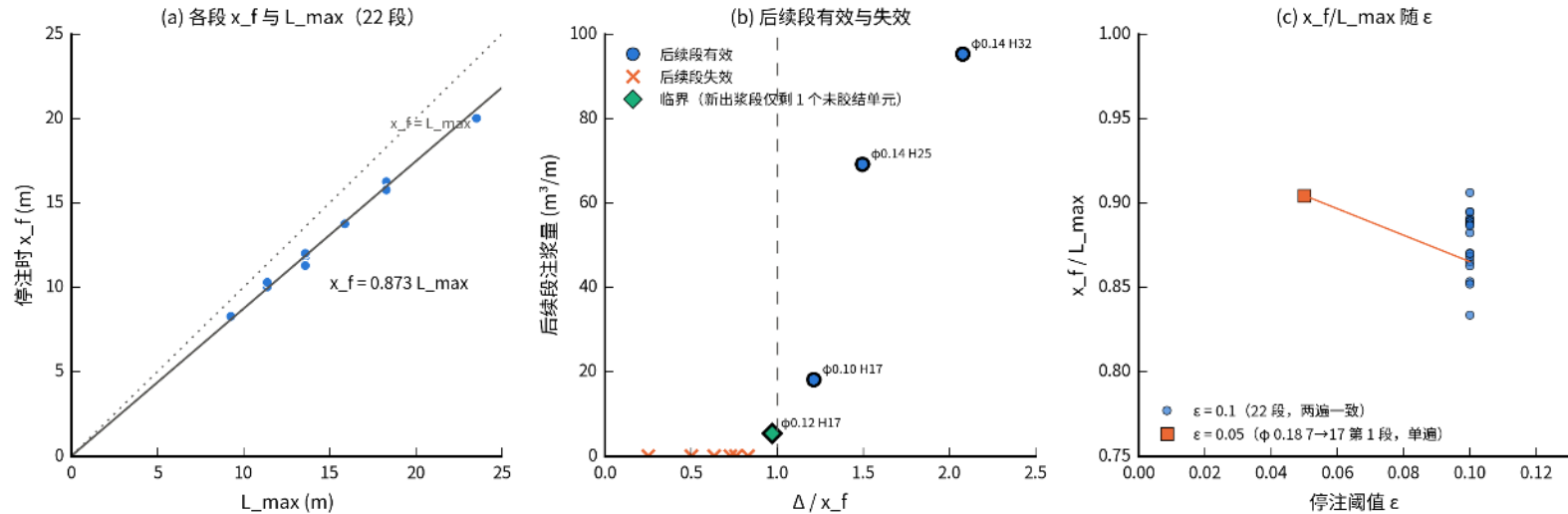


图 4.6-3

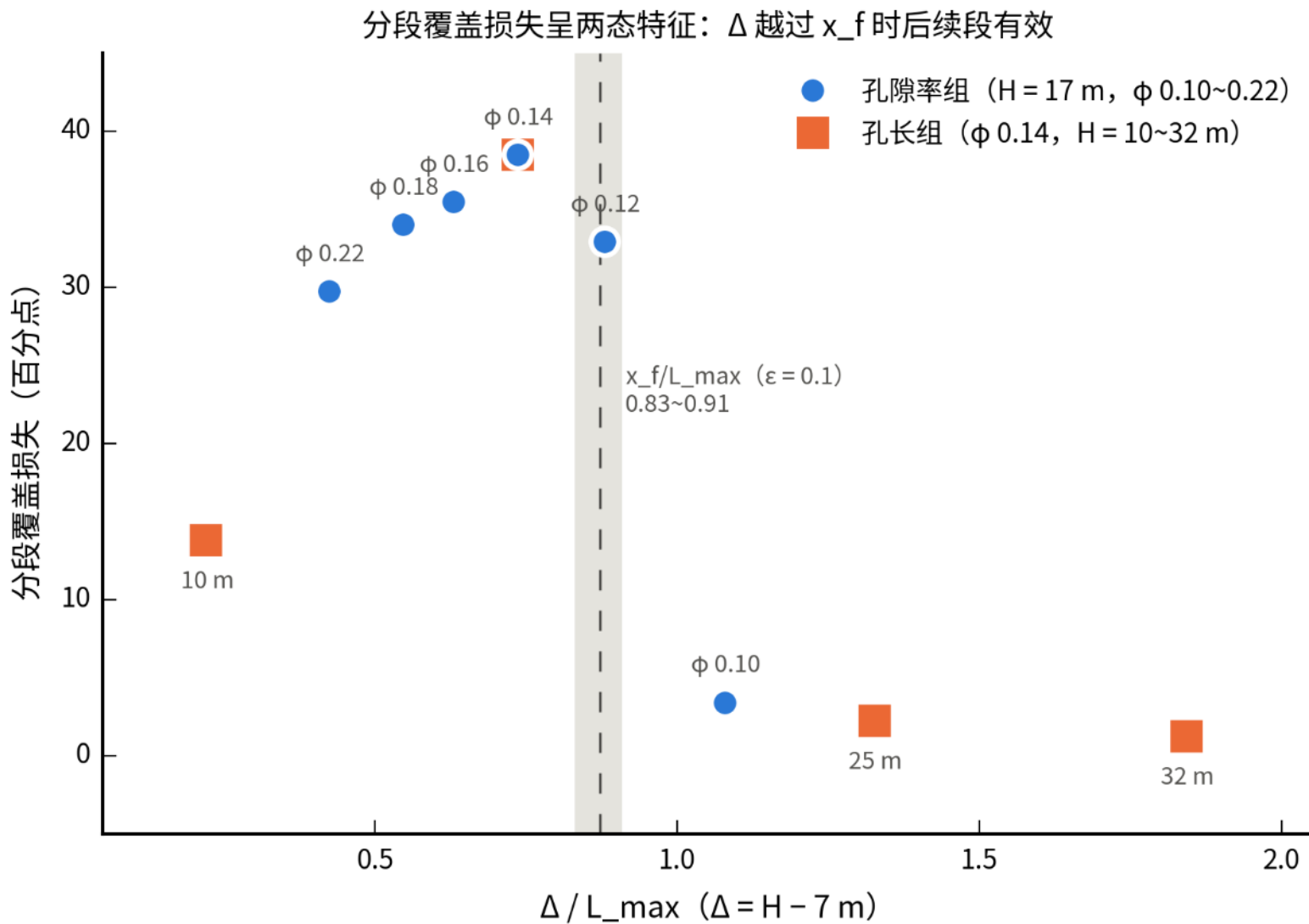


图 4.6-4

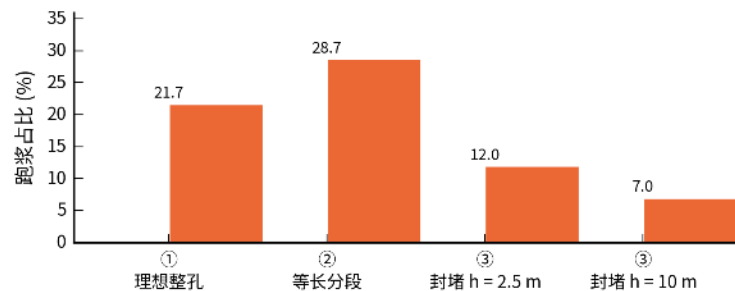
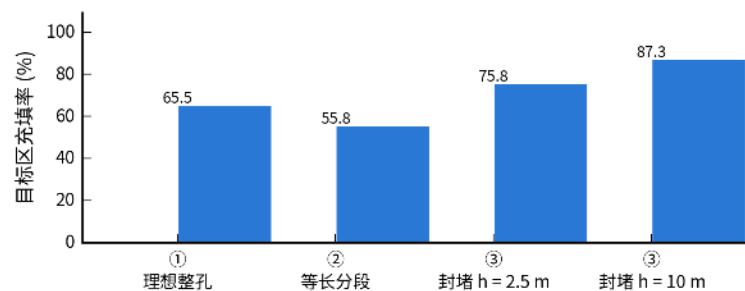
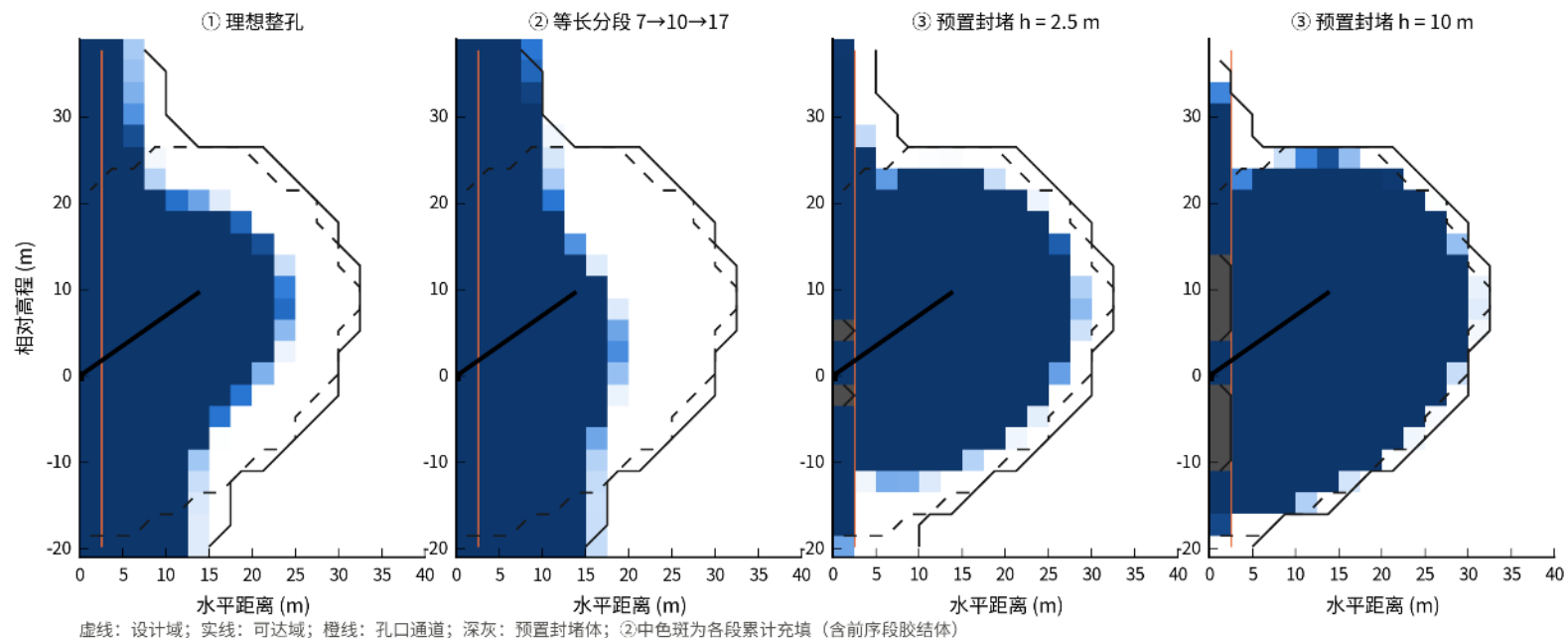
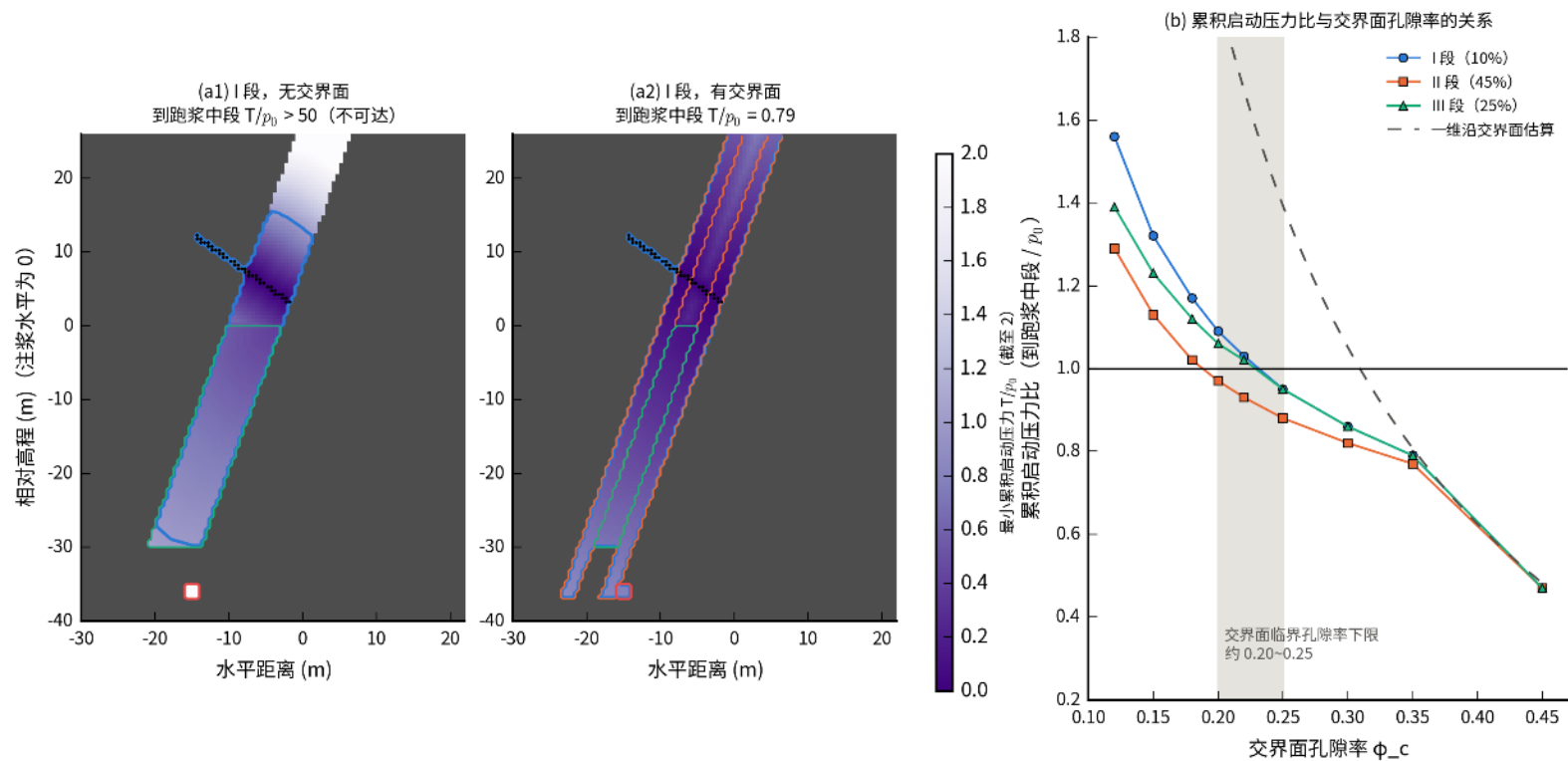
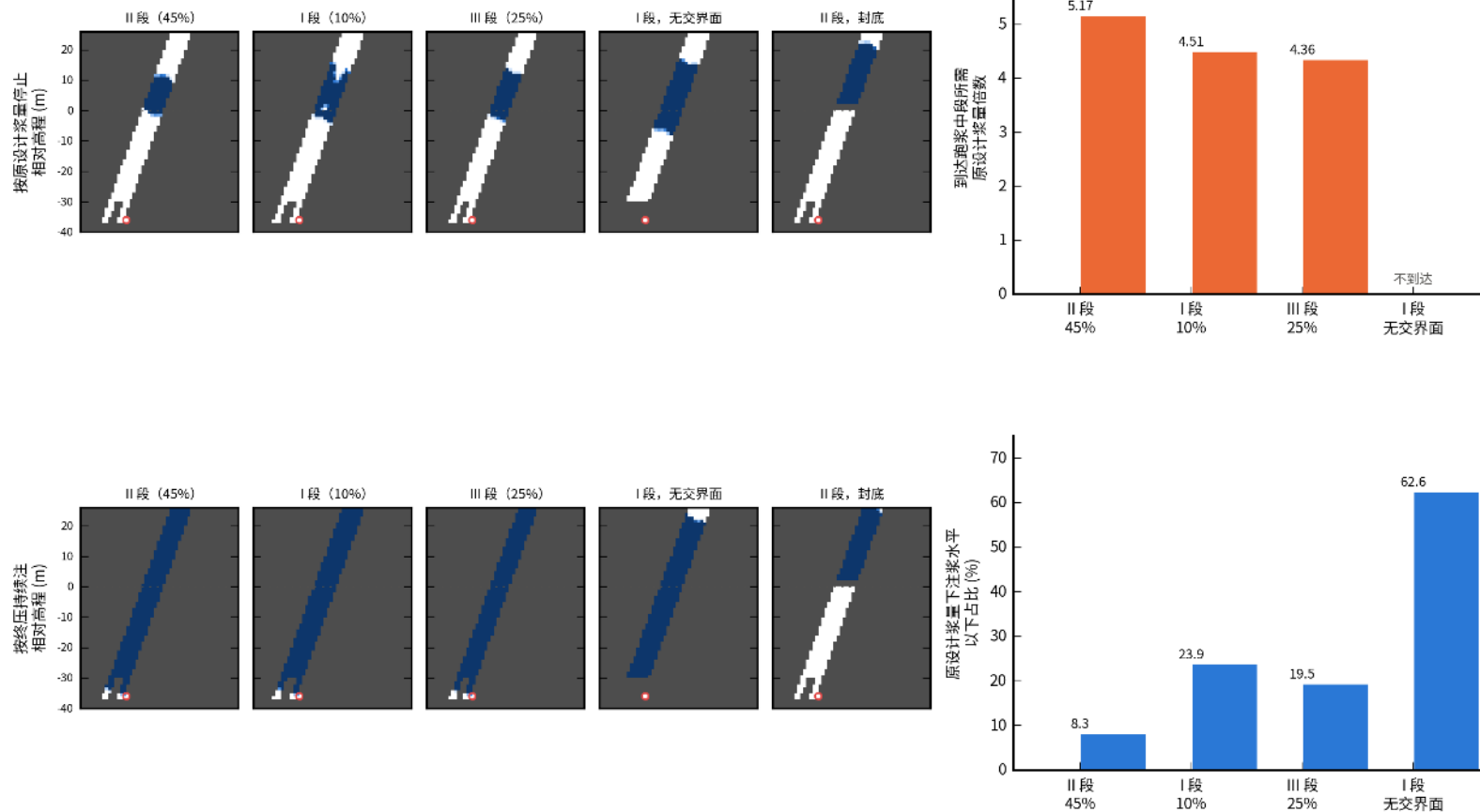


图 4.7-2



蓝线: 可达域边界 ($T = p_0$) ; 橙线: 界面; 绿线: 存窿矿石; 红框: 跑浆中段; 黑点: 钻孔出浆段; 深灰: 围岩与未采矿体 (不透浆)

图 4.7-3



色斑：充填域；深灰：围岩、未采矿体与封底；红框：跑浆中段；点线：注浆水平。上排E为封底后按变更设计浆量停止。